

Phys 115B HW 9

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Question 1. Consider the parity operator $\hat{\Pi}$ in three dimensions, such that $\hat{\Pi}\psi(\mathbf{r}) = \psi(-\mathbf{r})$.

(a) Show that for states ψ expressed in polar coordinates, the action of the parity operator can be written as

$$\hat{\Pi}\psi(r, \theta, \phi) = \psi(r, \pi - \theta, \phi + \pi)$$

(b) Show that for hydrogenic energy eigenstates $\psi_{nlm}(r, \theta, \phi)$ we have

$$\hat{\Pi}\psi_{nlm} = (-1)^l \psi_{nlm}(r, \theta, \phi)$$

Proof.

(a) Given the parity operator has the coordinate change $\mathbf{r} \mapsto -\mathbf{r}$, one has the following on each component:

$$x : r \sin(\theta) \cos(\phi) \mapsto -r \sin(\theta) \cos(\phi) = r \sin(\pi - \theta) \cos(\phi + \pi) \quad (1)$$

$$y : r \sin(\theta) \sin(\phi) \mapsto -r \sin(\theta) \sin(\phi) = -r \sin(\pi - \theta) \sin(\phi) = r \sin(\theta) \sin(\phi + \pi) \quad (2)$$

$$z : r \cos(\theta) \mapsto -r \cos(\theta) = r \cos(\pi - \theta) \quad (3)$$

So, one transformation on the spherical coordinates that's compatible with the parity operator is $r \mapsto r$, $\theta \mapsto \pi - \theta$, and $\phi \mapsto \phi + \pi$.

Hence, using r, θ, ϕ as the input, one has:

$$\hat{\Pi}\psi(r, \theta, \phi) = \psi(r, \pi - \theta, \phi + \pi) \quad (4)$$

(b) Since for $\psi_{nlm} = R_{n,l}(r)Y_l^m(\theta, \phi)$ in general, then the only one affected by the parity is $Y_l^m(\theta, \phi)$.

Recall the formula $Y_l^m(\theta, \phi) = e^{im\phi} P_l^m(\cos(\theta))$, where P_l^m is given as follow:

$$P_l^m(z) = (1 - z^2)^{|m|/2} \left(\frac{d}{dz} \right)^{|m|} P_l(z), \quad P_l(z) := \frac{1}{2^l l!} \left(\frac{d}{dz} \right)^l (z^2 - 1)^l \quad (5)$$

Which, it can be shown inductively, that all $P_l(z)$ only contains the monomials with the degree of same parity with l ; then, one can also inductively shown $\left(\frac{d}{dz} \right)^{|m|} P_l(z)$ has all monomial with degree of same

parity with $l - |m|$, showing $P_l^m(-z) = (-1)^{l-|m|}P_l^m(z)$. Finally, one has $e^{im(\phi+\pi)} = (-1)^{|m|}e^{im\phi}$. Hence, with $\cos(\pi - \theta) = -\cos(\theta)$, one has the following:

$$\hat{\Pi}\psi_{nlm} = \psi_{nlm}(r, \pi - \theta, \phi + \pi) = R_{n,l}(r)e^{im(\phi+\pi)}P_l^m(\cos(\pi - \theta)) \quad (6)$$

$$= R_{n,l}(r)(-1)^{|m|}e^{im\phi}P_l^m(-\cos(\theta)) = R_{n,l}(-1)^{|m|}e^{im\phi}(-1)^{l-|m|}P_l^m(\cos(\theta)) \quad (7)$$

$$= (-1)^l R_{n,l}e^{im\phi}P_l^m(\cos(\theta)) = (-1)^l \psi_{nlm}(r, \theta, \phi) \quad (8)$$

□

Question 2. We can classify operators depending on how they transform under parity, $\hat{Q} \rightarrow \hat{\Pi}^\dagger \hat{Q} \hat{\Pi}$ (see example 6.2 or Table 6.1 in the book can be useful).

(a) In one dimension, show that position and momentum are odd under parity.

(b) In three dimensions, operators that are invariant under parity and commute with $\hat{L}$ are called scalar operators, $\hat{\Pi}^\dagger \hat{Q}_s \hat{\Pi} = \hat{Q}_s$. Operators that flip sign under parity and commute with $\hat{L}$ are called pseudo-scalar operators, $\hat{\Pi}^\dagger \hat{Q}_s \hat{\Pi} = -\hat{Q}_s$. Show that scalar operators satisfy $[\hat{\Pi}, \hat{Q}_s] = 0$, while pseudoscalar operators satisfy

$$\{\hat{\Pi}, \hat{Q}_s\} = 0$$

(where the curly brackets denote the anti-commutator).

(c) Operators that flip sign under parity and whose commutators with $\hat{L}$ are like those of $\hat{L}$ are called vector operators, such that $\hat{\Pi}^\dagger \hat{Q}_v \hat{\Pi} = -\hat{Q}_v$. Operators that are invariant under parity and whose commutators with $\hat{L}$ are like those of $\hat{L}$ are called pseudovector or axial vector operators, $\hat{\Pi}^\dagger \hat{Q}_a \hat{\Pi} = \hat{Q}_a$. Show that vector operators satisfy

$$\{\hat{\Pi}, \hat{Q}_v\} = 0$$

and pseudo-vectors satisfy

$$[\hat{\Pi}, \hat{Q}_a] = 0.$$

(d) Give an example of a scalar operator, pseudoscalar operator, a vector operator and an axial or pseudo vector operator (feel free to Google or use the book).

Proof.

(a) Let's first recall some tools, for instance $\langle x|p\rangle = e^{ipx/\hbar}$. Then, we have the following:

$$\hat{P}^\dagger \hat{x} \hat{\Pi} |x\rangle = \hat{\Pi} \hat{x} | -x\rangle = -x \hat{P} | -x\rangle = -x |x\rangle \quad (9)$$

Since above is true for all $x \in \mathbb{R}$, it shows $\hat{\Pi}^\dagger \hat{x} \hat{\Pi} = -\hat{x}$.

Similarly, we have the following:

$$\hat{\Pi} |p\rangle = \int \hat{\Pi} |x\rangle \langle x|p\rangle dx = \int e^{ipx/\hbar} | -x\rangle dx \quad (10)$$

$$= \int e^{i(-p)u/\hbar} |u\rangle du = | -p\rangle \quad (11)$$

(Note: Here $u := -x$).

Hence, the full relation on momentum is:

$$\hat{\Pi}^\dagger \hat{p} \hat{\Pi} |p\rangle = \hat{\Pi} \hat{p} | -p\rangle = -p \hat{\Pi} | -p\rangle = -p |p\rangle \quad (12)$$

Since above is true for all $p \in \mathbb{R}$, it shows $\hat{\Pi}^\dagger \hat{p} \hat{\Pi} = -\hat{p}$.

(b) For scalar operators we have the following:

$$\hat{\Pi}^\dagger \hat{Q}_s \hat{\Pi} = \hat{Q}_s \implies \hat{Q}_s \hat{\Pi} = \hat{\Pi} \hat{Q}_s \implies [\hat{\Pi}, \hat{Q}_s] = 0 \quad (13)$$

For pseudo-scalar operators we have the following:

$$\hat{\Pi}^\dagger \hat{Q}_s \hat{\Pi} = -\hat{Q}_s \implies \hat{Q}_s \hat{\Pi} = -\hat{\Pi} \hat{Q}_s \implies \{\hat{\Pi}, \hat{Q}_s\} = 0 \quad (14)$$

(c)

□

Question 3. Consider a particle of mass m and charge q confined to a circular ring in the x - y plane, with radius r . The system will be driven by an electric field E_x in the x direction (which can stimulate photon induced transitions). It is natural to work in polar coordinates for this problem, with $x = r \cos \theta$.

- (a) What are the first five energies of the system with $E_x = 0$?
- (b) What are the first five eigenfunctions $\psi(\theta)$? Please normalize them, and pick your basis such that the eigenfunctions are also eigenfunctions of L_z .
- (c) Explicitly write out a 3×3 matrix for the Hamiltonian

$$H' = E_x q x = E q r \cos \theta$$

This matrix will use the three lowest states above. Extrapolate this result to write a concise selection rule that tells what transitions are allowed for any state.

- (d) What frequencies are absorbed by a “gas” of such rings if they are cold enough to be initially in the ground state?
- (e) What additional frequencies are absorbed by rings initially in the first excited state?

Proof.

□